



Beyond digital finance: The impact of internet banking adoption on subjective life satisfaction

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ABSTRACT

The rapid development of digital finance has made Internet banking a key part of modern financial services, shaping consumer behavior and daily life. The rise of purely Internet-based banks (PIBs) operating entirely on digital platforms has further changed traditional banking. However, their impact on consumer behavior and subjective life satisfaction remains unclear. Most prior studies rely on cross-sectional data, limiting understanding of pre- and post-PIB introduction dynamics. Additionally, the roles of technological well-being and e-commerce participation in shaping perceived ease of use and perceived usefulness are underexplored. Furthermore, the broader impact of digital finance on subjective life satisfaction has not been well studied. This study uses data from Taiwan's Digital Development Survey (2019, 2021, and 2022) to explore key factors influencing Internet banking adoption with a Logit model, examine its link to subjective life satisfaction using a Tobit model, and assess the policy impact of PIBs through the Difference-in-Differences (DID) method. Empirical results show that perceived usefulness (PU) and perceived ease of use (PEU) significantly influence Internet banking adoption. Technological well-being enhances PEU, while participation in e-commerce strengthens PU. The strong link between behavioral intention and actual usage confirms the relevance of the Technology Acceptance Model (TAM) in digital finance. The initial introduction of PIBs had a negative impact on subjective life satisfaction, likely due to adaptation costs and perceived risks. However, this effect faded within a year as satisfaction levels recovered. This study is the first to integrate the TAM framework with the DID approach, providing empirical evidence on PIBs' effects on consumer behavior and satisfaction. The findings suggest that governments should promote digital financial education to close the digital divide, while financial institutions should enhance user experience and security to support market adaptation.

1. Introduction

The rapid growth of digital finance is reshaping global financial markets and consumer behavior. Internet banking and purely Internet-based banks (PIBs) have significantly affected traditional banking (Ren et al., 2022). By offering online services such as deposits, transfers, loans, and investment management, Internet banking has reduced dependence on physical branches, improving transaction efficiency and user convenience. However, adoption rates vary across countries and demographic groups, as some users hesitate due to concerns about trust, privacy, cybersecurity risks, and technological adaptability (Bai et al., 2022).

Internet banking has evolved over the past two decades in Taiwan through concerted industry and government efforts. The Ministry of Finance began promoting online banking as early as the late 1990s, implementing policies to foster Internet banking development. As a

result, by the 2010s, virtually all major banks in Taiwan were offering Internet and mobile banking services, and consumers had grown accustomed to the convenience of online financial transactions. This widespread uptake of Internet banking services provided a strong foundation for further financial innovation. 2018 Taiwan's Financial Supervisory Commission (FSC) introduced a framework to license Internet-only banks as a new fintech initiative. Three consortiums were approved in 2019 (each including incumbent financial institutions as required by regulators) to set up the first generation of purely internet-based banks, paving the way for their official market entry by 2020–2021.

Taiwan's approval of three PIBs in 2021 marked a significant step in its digital financial reform. The three new entrants – Rakuten International Commercial Bank, LINE Bank, and Next Bank – launched between 2020 and 2022 under this policy-driven reform. Rakuten Bank, backed by Japan's Rakuten Group, was the first to commence operations

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(January 2021). It leverages its parent company's e-commerce ecosystem to attract users with integrated loyalty rewards and services. LINE Bank, which went live in April 2021 as a subsidiary of the popular LINE messaging platform, similarly capitalizes on its extensive user base and existing digital payment (Line Pay) infrastructure to offer socially integrated financial products. Next Bank, backed by Chunghwa Telecom alongside financial partners, launched in early 2022 after a more extended preparatory period. Its telecom affiliation represents a convergence of telecommunications and banking, positioning it to explore unique cross-industry services. Unlike traditional Internet banking, PIBs operate without physical branches, relying entirely on digital technologies and artificial intelligence (AI). While adoption is increasing, their impact on consumer behavior and subjective life satisfaction remains unclear. This study investigates the key determinants of Internet banking adoption in Taiwan and evaluates whether PIBs have influenced consumer usage behavior and life satisfaction.

A review of existing studies shows that most research on Internet banking adoption is based on the Technology Acceptance Model (TAM) (Yu, 2012; Ho et al., 2020). These studies consistently identify perceived usefulness (PU) and perceived ease of use (PEU) as key drivers of consumer intentions (Liébana-Cabanillas et al., 2018; Bekmukhambetova & Németh, 2023). However, three research gaps remain. First, most studies use cross-sectional data, leaving unclear how consumer behavior and satisfaction evolved before and after the introduction of PIBs. Second, while PU and PEU are widely recognized, the roles of technological well-being and e-commerce participation in shaping these factors remain unexamined. Third, although digital finance is known for its convenience and efficiency, its broader impact on subjective life satisfaction has yet to be fully explored.

To address these gaps, this study applies a Tobit model to examine the relationship between Internet banking usage and life satisfaction. It employs the Difference-in-Differences (DID) method to evaluate the policy and social effects of PIBs adoption systematically. The research has three objectives. First, it identifies the main factors influencing Internet banking usage in Taiwan and assesses TAM's applicability in a digital finance setting. Second, it applies the DID model to measure the impact of PIBs on consumer behavior and life satisfaction. Third, it explores how digital environmental factors—particularly technological well-being and e-commerce participation—interact with TAM, enhancing its explanatory power in a highly digitalized financial landscape.

This study makes several contributions. It examines consumer behavior and life satisfaction before and after PIBs implementation, addressing gaps in policy-related research. It also expands TAM by integrating technological well-being and e-commerce participation, making the model more relevant to fintech developments. Finally, assessing the long-term effects of Internet banking on life satisfaction provides insights for policymakers and financial institutions seeking to promote inclusivity and maximize societal benefits in digital finance.

This study uses digital behavior survey data from Taiwan's Ministry of Digital Affairs, collected in 2019, 2021, and 2022. Three econometric models—Logit, Tobit, and Difference-in-Differences—are applied to test the research hypotheses. The Logit model identifies key factors influencing electronic payment intentions and Internet banking usage, evaluating the applicability of the core TAM constructs: PU, PEU, and behavioral intention. The Tobit model assesses whether Internet banking usage improves subjective life satisfaction. The DID model examines how PIBs affect consumer behavior and life quality during the market adaptation phase.

The study's hypothesis framework is based on TAM, incorporating technological well-being and e-commerce participation as factors influencing PEU and PU, respectively. It also explores how PEU and PU shape behavioral intention and how behavioral intention and actual usage impact subjective life satisfaction. Additionally, the DID model evaluates the policy effects of PIBs implementation. The findings

provide insights into digital financial reforms, offer empirical evidence on PIBs adoption, and inform policymakers and financial institutions in developing future Internet banking strategies.

The study is structured as follows. Section 2 reviews the literature on digital finance, the Technology Acceptance Model, PIBs development, and key factors affecting Internet banking adoption. Section 3 describes the data sources. Section 4 introduces the theoretical framework and empirical methodology, detailing the applications of the Logit, Tobit, and DID models. Section 5 presents the empirical results. Section 6 discusses findings about existing literature, examines policy implications and research limitations, and provides recommendations for government regulators and financial institutions. This section also summarizes key contributions and suggests directions for future research.

2. Literature review

The rapid growth of the digital economy has reshaped financial market operations and financial service delivery. Ren et al. (2022) note that the digital economy fosters new interactions between industries and financial markets beyond altering global economic structures, creating spillover effects within financial systems. This shift has accelerated the adoption of digital financial services, including Internet and mobile banking. From a supply chain finance perspective, Bai et al. (2022) point out that while blockchain technology improves transaction transparency and efficiency through decentralization and immutability, challenges such as weak legal frameworks, limited cross-chain interoperability, and poor data governance hinder its full integration into digital financial services.

These macroeconomic and technological factors indicate that Internet banking expansion is not solely driven by consumer demand but also depends on regulatory conditions and technological readiness. Promoting Internet banking, therefore, requires a careful approach. Mansumittrchai and Chiu (2012) provide evidence from the United Arab Emirates (UAE), showing that weak regulatory and technological infrastructure and consumers' preference for face-to-face interactions significantly hinder online banking adoption.

2.1. Theoretical perspectives on internet banking adoption

Research on Internet and mobile banking adoption often integrates multiple theoretical frameworks, including the Technology Acceptance Model (TAM), Diffusion of Innovation (DOI), and the Theory of Planned Behavior (TPB) or its decomposed variant (DTPB) (Yu, 2012; Ho et al., 2020). These models incorporate core constructs such as perceived usefulness (PU) and perceived ease of use (PEU) while extending them with factors like trust, risk perception, security, and privacy concerns (Liébana-Cabanillas et al., 2018; Shareef et al., 2018; Souiden et al., 2021). This integration provides a more comprehensive understanding of digital financial service adoption.

Empirical studies further support these frameworks. Chandio et al. (2013) found that technological self-efficacy and trust strengthen PU and PEU in predicting Internet banking adoption in Pakistan. Alalwan et al. (2017) similarly emphasized trust as a key factor in mobile banking adoption in developing countries. Bekmukhambetova and Németh (2023) reaffirmed TAM's predictive power in Internet banking, highlighting the strong influence of PU and PEU. Hernandez and Mazzon (2007) argued for broader integration of technology adoption models, showing that compatibility, self-efficacy, and external facilitating conditions collectively shape behavioral intention and actual usage. These findings reinforce the role of external resources and individual capabilities in driving Internet banking adoption.

2.2. The role of DOI and risk perception in Internet banking

According to the DOI theory, compatibility and relative advantage

are key factors driving Internet banking adoption, while complexity is a significant barrier. Ndubisi and Sinti (2006) and Tan and Teo (2000) confirmed these effects in studies on Malaysia and Singapore. To deepen the understanding of user behavior, TPB and DTPB are often integrated with TAM or DOI (Lee, 2009; Danyali, 2018), allowing for a more detailed analysis of how constructs like attitude, subjective norms, and perceived behavioral control link user intention to actual behavior.

Lee (2009) expanded this framework by identifying five dimensions of perceived risk: security, financial, time, performance, and social risks. His findings suggest that each type of risk influences Internet banking adoption differently, offering a more nuanced perspective for future research. Similarly, Sharma et al. (2020) applied the Unified Theory of Acceptance and Use of Technology (UTAUT) in a study on Fiji, incorporating perceived risk and customer satisfaction as key factors. Their results show that cultural differences shape adoption intentions, emphasizing the need for banks to allocate resources strategically and enhance security measures to reduce risk concerns, especially in developing economies.

2.3. Cross-cultural adoption studies

Cross-cultural and regional studies have validated TAM across different economies, identifying key factors influencing mobile banking adoption. Malaquias and Hwang (2019) highlighted the importance of trust and social influence in developing countries such as Brazil. In India, studies by Bashir and Madhavaiah (2015), Bhatt (2016), and Inder et al. (2022) consistently identified security concerns, technological barriers, and low awareness as significant obstacles to adoption. Research in Iran (Danyali, 2018) and Vietnam (Lin et al., 2015) further confirmed that compatibility, subjective norms, and resource support play crucial roles in shaping adoption behavior. Studies in Malaysia and Ethiopia (Angusamy et al., 2022; Ayinaddis et al., 2023) emphasized the role of reliability and security in customer satisfaction and loyalty, reinforcing the importance of service quality in adoption decisions.

In China, Zhan and Huang (2019) found that financial benefits, PU, and platform convenience positively influence Internet banking usage frequency and transaction amounts. Wang et al. (2003) showed that perceived credibility has a more substantial impact on behavioral intention than PU among Taiwanese users. Ho et al. (2020) combined DOI, TAM, and DTPB to develop a cross-cultural framework for mobile banking adoption. Their comparison of Taiwan and Vietnam showed that while trialability and compatibility vary by context, PU and PEU remain the main drivers of adoption.

In Western countries, large-scale longitudinal studies by Furst et al. (2002) and Xue et al. (2011) analyzed how Internet banking affects transaction behavior and customer loyalty. Their findings indicate that users with higher transaction needs are more likely to increase their use of digital services. In contrast, those with higher education levels or greater financial literacy show stronger adoption tendencies. In the UK, Yousafzai and Yani-de-Soriano (2012) incorporated technological readiness and demographics, revealing that young males prioritize PU while older females emphasize PEU. Differences in technological readiness also shape perceptions of risk and security.

Estrella-Ramon et al. (2016) found that offline transaction behavior, customer loyalty, and cross-product purchases further drive Internet banking adoption in Spain. In Italy, Capece and Campisi (2013) showed that service quality, information quality, and trust influence user satisfaction and adoption more than ease of use. Studies in Northern Cyprus and Oman (Alhassany & Faisal, 2018; Sharma & Sharma, 2019) pointed to website usability and trust as key adoption factors, while security and privacy concerns remained significant barriers.

In other regions, such as Cyprus (Andreou & Anyfantaki, 2021) and Greece (Patsiotis et al., 2012; Santouridis & Kyritsi, 2014), Internet banking acceptance varies significantly, particularly in the context of financial crises or low financial literacy. In these areas, non-adopters face dual challenges—high-risk perception and limited access to

financial and technological resources—further deepening the digital divide in banking services.

2.4. Trust, risk, and financial literacy in digital finance

Existing research highlights trust and risk perception as fundamental factors in digital financial service adoption (Alalwan et al., 2017; Ho et al., 2020). Lee (2009) identified security and financial risks as the most significant barriers. Aboobucker and Bao (2018) found that a user-friendly and secure website interface in Sri Lanka significantly reduces privacy concerns. Hu et al. (2019) further emphasized that trust is a key mediating factor in China, where government support, brand reputation, and user innovativeness influence behavioral intentions through trust. As emerging technologies like blockchain (Bai et al., 2022) and purely internet-based banks (Lee & Kim, 2020; Kavaya & Rakesh, 2022) gain traction, security assurances, and risk management have become increasingly critical in building trust and encouraging adoption.

Beyond technological and trust-related factors, financial literacy and digital capability are primary in adoption patterns. Andreou and Anyfantaki (2021) found that individuals with lower financial knowledge are more likely to be discouraged by perceived risks in Internet banking. Similarly, Yu (2012) and Bekmukhambetova and Németh (2023) showed that gender and age moderate the effects of PEU and technological self-efficacy on adoption behavior. Grigorescu et al. (2023), in an analysis of 27 EU countries, revealed that individuals with higher education levels and middle-aged users exhibit significantly higher adoption rates than older or socioeconomically disadvantaged groups. This finding underscores the importance of digital inclusion, emphasizing the need for financial institutions and policymakers to develop services tailored to diverse demographic needs.

Widayat et al. (2020) examined Indonesian adolescents' adoption of electronic payments using TPB, TAM, and UTAUT models in emerging markets. Their findings highlight the role of social influence and effort expectancy in shaping adoption intentions, reinforcing the need for infrastructure development and educational outreach to expand access to digital financial services.

2.5. Longitudinal and two-stage approaches

Recent studies increasingly adopt a two-stage approach to analyze Internet and mobile banking behavior, distinguishing between pre- and post-adoption phases (Montazemi & Qahri-Saremi, 2015; Shareef et al., 2018). This method clarifies the difference between initial acceptance and sustained usage. Sharma and Sharma (2019) found that service quality and system reliability are key factors in continued Internet banking use. Similarly, Xue et al. (2011), using survival analysis in the U.S., showed that adopters tend to increase transaction frequency and loyalty over time. However, banks lacking sufficient scale or technological capability may face short-term challenges due to high investment costs and cybersecurity risks (Furst et al., 2002). External shocks, such as digital economy growth (Ren et al., 2022) and the COVID-19 pandemic (Inder et al., 2022), have introduced fluctuations in digital financial adoption, highlighting consumer behavior's evolving and dynamic nature. These findings emphasize the importance of a longitudinal perspective in assessing the long-term sustainability of Internet banking services.

As traditional banks accelerate digital transformation, PIBs and emerging financial platforms continue to attract users, making network externalities increasingly relevant in adoption behavior. In South Korea, Lee and Kim (2020) found that reaching a critical mass of key users and offering diverse services are crucial in the early phase, while economic benefits and security risks shape continued usage. Kavaya and Rakesh (2022) noted that Internet banking is valued for transaction convenience and processing speed, though technical issues and security concerns remain significant barriers. Earlier research by Furst et al. (2002) in the

U.S. indicated that electronic banking enhances customer loyalty and income growth within specific demographic groups. However, small and mid-sized banks face uncertainties in managing capital investments and risk controls for large-scale Internet banking expansion. These findings underscore the need for banks to balance service innovation, user engagement, and risk management to ensure sustainable digital financial growth.

2.6. Integrated insights on TAM, well-being, and demographic factors

The literature consistently underscores the centrality of the Technology Acceptance Model (TAM) in explaining Internet banking adoption. In particular, TAM's two core constructs—perceived usefulness (PU) and perceived ease of use (PEOU)—emerge as universally significant predictors of user acceptance of Internet banking services. Numerous studies across different contexts reaffirm that when users believe an online banking platform is helpful for their needs and easy to operate, they are far more likely to intend to use and adopt it. Even when additional factors (e.g., trialability, compatibility, or cultural nuances) are introduced, PU and PEOU remain the primary drivers of adoption behavior. This robust finding validates TAM as a foundational theoretical framework for understanding consumer behavior in digital finance.

Beyond TAM's core, researchers have integrated complementary variables to enrich the explanation of adoption. Trust in the service provider and platform is often highlighted as a critical enabler, whereas perceived risk—especially security and privacy concerns—is a common barrier. Studies show that strong security assurances and a trustworthy reputation encourage online banking usage, whereas fear of fraud or privacy breaches deters users. These insights have led to extended TAM models incorporating trust and risk as external variables influencing PU and PEOU or directly affecting usage intentions.

Social influence—including normative pressure or peer effects—also shapes attitudes toward fintech services, particularly within certain demographic groups or cultures. For example, evidence from emerging markets shows that peer and family influence can stimulate e-payment adoption among young users (Widayat et al., 2020). Such findings align with extensions of TAM via theories like the Theory of Planned Behavior and UTAUT, which add social norms as determinants of technology adoption. The literature affirms that while many factors can affect Internet banking uptake, ease-of-use, and usefulness perceptions remain fundamental, and trust-related constructs frequently moderate these perceptions in practice.

An emerging stream of inquiry—and a key focus of this study—is the link between Internet banking adoption and user well-being outcomes. *Subjective life satisfaction* refers to an individual's overall assessment of their quality of life or happiness. In the Internet banking context, it represents how Internet banking might improve (or impair) an individual's well-being—for example, by saving time, reducing stress, or easing financial management. However, the literature on this relationship remains sparse: most technology adoption studies stop at behavioral intention or actual usage and do not track broader life-quality impacts. This gap highlights an opportunity to explore whether increased fintech use yields tangible improvements beyond immediate service utility.

Closely related is technological well-being, which captures an individual's capacity to thrive in a technology-driven environment. It combines digital literacy, reliable access, and the ability to integrate digital tools into daily life. Individuals with higher technological well-being will likely view new fintech services as less intimidating and more approachable, indirectly enhancing PEOU. Although prior studies touch on concepts such as digital literacy and technological self-efficacy, technological well-being as a unified measure has been under-explored—an innovative angle the present study adopts to extend TAM's explanatory power.

Demographic characteristics further contextualize these behavioral factors. Age, gender, and education level have been shown to influence

Internet banking usage patterns systematically: younger and more-educated individuals adopt such services more readily than older or less-educated cohorts. Gender effects have also been documented—some research finds women at least as likely if not more so, to embrace online financial services, often linked to differing financial management roles or trust perceptions. Highlighting these demographics is vital for digital inclusion debates: if certain groups (e.g., seniors, rural residents, or the less educated) lag in adoption, targeted strategies are needed to address their specific barriers.

That said, prior studies often treat demographics as control variables rather than core theoretical constructs, resulting in a less cohesive understanding of how and why different segments adopt (or avoid) Internet banking. Few works examine demographic-driven differences in TAM constructs or long-term well-being outcomes. This limitation justifies the present study's deeper focus on demographic heterogeneity. By integrating life-satisfaction outcomes and explicit considerations of technological well-being and user demographics, the current research extends the TAM framework, illuminating both the adoption decision process and its broader implications for diverse segments of society.

Table 1 below presents a summary of the key variables identified in the literature, including their definitions, theoretical foundations, and illustrative references drawn from previous studies:

This consolidated view highlights how TAM's core constructs (PU, PEOU) operate with other influential variables. It also stresses the importance of considering user preparedness (technological well-being), outcome benefits (life satisfaction), and user heterogeneity (demographics) when examining Internet banking adoption. By organizing the literature in this way, this study emphasizes the critical roles of PU and PEOU alongside complementary factors, and we identify where gaps—such as the technology–well-being linkage and demographic disparities—remain to be explored. This sets the stage for the subsequent analysis, where the theoretical model (Section 4) and empirical investigation will build upon these insights to advance a more comprehensive understanding of Internet banking adoption and its implications.

3. Data

This study uses data from the Digital Development Survey (DDS) conducted by Taiwan's Ministry of Digital Affairs (MODA). This nationally representative longitudinal dataset tracks public engagement with digital technologies, internet applications, digital services, and attitudes toward technological development. Since its launch in 2001, the survey has been conducted annually, except in 2020, when data collection was suspended due to the COVID-19 pandemic. The questionnaire is periodically revised to reflect changes in digital technology and societal behavior, ensuring the data remains relevant. These design features provide strong continuity and temporal validity, making the dataset well-suited for individual-level analysis.

The study's primary variable of interest comes from the survey question: "Have you used Internet or mobile banking in the past three months?" Respondents answered "Yes" or "No," allowing the construction of a binary dependent variable. To assess the impact of PIBs introduction on Internet banking adoption before, during, and after its launch, the study uses survey data from 2019, 2021, and 2022, corresponding to survey publications in 2020, 2022, and 2023. After excluding missing values, the final dataset includes 11,708 observations for 2020, 10,427 for 2022, and 10,916 for 2023, providing a robust sample size for cross-year empirical analysis.

By leveraging this representative longitudinal dataset, the study offers a comprehensive view of the evolving digital behavior of Taiwanese citizens. It empirically examines the impact of PIBs introduction on Internet banking usage. The findings contribute to academic research and policy discussions on digital financial service development.

Table 1
Principal constructs, definitions, theoretical foundations, and representative references in Internet-banking adoption research

Construct / Variable	Definition (abridged)	Theoretical Foundation	Representative References
Perceived Usefulness (PU)	The degree to which Internet banking service is perceived to provide value or enhance the user's banking performance (e.g., efficiency, convenience, effectiveness).	Technology Acceptance Model (TAM) – core belief influencing attitude/intention.	Ho et al. (2020); Liébana-Cabanillas et al. (2018)
Perceived Ease of Use (PEOU)	The degree to which the service is perceived as free of effort or difficulty. In online banking, the user finds the platform intuitive and easy to learn or operate.	Technology Acceptance Model (TAM) – core belief influencing attitude/intention.	Liébana-Cabanillas et al. (2018); Bekmukhambetova & Németh (2023)
Trust (in technology/service)	User's confidence that the Internet banking platform is secure and reliable, and that the provider will safeguard user interests (e.g., protecting privacy, ensuring transactions are safe).	Extended TAM & E-Commerce Trust Models – trust as an external factor boosting acceptance by reducing perceived risk.	Hu et al. (2019); Lee & Kim (2020)
Perceived Risk (security/privacy concerns)	The degree to which a user anticipates potential negative consequences from online banking, such as financial loss, fraud, or privacy invasion. High perceived risk undermines adoption.	Perceived Risk Theory in online services; incorporated in TAM extensions and UTAUT as a barrier to adoption.	Lee (2009); Aboobucker & Bao (2018)
Social Influence (subjective norm)	The impact of others' opinions and usage on an individual's decision to adopt. For instance, recommendations or usage by friends, family, or peers can encourage one to try Internet banking.	Theory of Planned Behavior; UTAUT – inclusion of social norms/influence affecting behavioral intention.	Widayat et al. (2020); Souiden et al. (2021)
Financial Literacy	The level of a person's financial knowledge and understanding of financial products affects their ability to use online banking confidently. Low financial literacy can heighten risk perceptions.	Consumer Financial Behavior and Inclusion – often a control/moderator in adoption studies linking knowledge to lower perceived complexity/risk.	Andreou & Anyfantaki (2021); Grigorescu et al. (2023)
Technological Well-Being (digital literacy)	An individual's overall digital capability and comfort – encompassing access to	Extended TAM (proxy for PEOU) – concept aligns with digital inclusion	Bekmukhambetova & Németh (2023); Chandio et al. (2013)

Table 1 (continued)

Construct / Variable	Definition (abridged)	Theoretical Foundation	Representative References
	technology, digital skills, and integration of tech into daily life. High-tech well-being implies that the user easily adapts to new online services.	literature; higher tech well-being lowers perceived effort in adoption.	
Life Satisfaction (subjective well-being)	A broad measure of overall contentment and quality of life. In this context, it reflects whether Internet banking correlates with improvements in an individual's life or happiness.	Subjective Well-Being theory (Psychology/Economics) – often used as an outcome in well-being studies; here considered as a long-term impact of technology use.	Limited prior research directly linking IB adoption to life satisfaction (noted as a gap)
Demographic Factors (Age, Gender, Education, etc.)	User background characteristics that can influence adoption. For example, age may reflect tech familiarity, and education can relate to tech and financial literacy; gender roles may affect trust or usage patterns.	Diffusion of Innovation & Digital Divide perspectives – demographics often moderate TAM/UTAUT effects and indicate inclusion gaps.	Yu (2012); Yousafzai & Yani-de-Soriano (2012); Grigorescu et al. (2023)

4. Model

This study is based on the Technology Acceptance Model and incorporates elements from Behavioral Utility Theory to construct a theoretical framework and empirical model. It examines how Internet banking usage affects subjective life satisfaction, focusing on how technology adoption influences individual utility and, in turn, enhances well-being. By integrating policy intervention analysis, the study also evaluates the dynamic effects of PIBs introduction before and after its implementation. This approach deepens theoretical understanding while providing empirical insights into the broader implications of digital financial services on individual well-being.

Assume that the banking service-related utility of individual *i* at time *t* is derived from individual characteristics (*Z_{it}*) and internet banking usage behavior (*IntBankUse_{it}*). The utility function can be expressed as:

$$U_{it} = U(Z_{it}, IntBankUse_{it}) \tag{1}$$

Internet banking usage influences utility through two main pathways. First, the transaction cost reduction effect (*TC_{it}*), wherein internet banking shortens transaction time and lowers economic costs. Second, the resource allocation efficiency enhancement effect (*Eff_{it}*), where Internet banking improves financial flexibility by offering convenient financial management and payment services. Consequently, individuals' decision to use Internet banking services depends on the utility maximization condition:

$$\Delta U = U(Z_{it}, IntBankUse_{it} = 1) - U(Z_{it}, IntBankUse_{it} = 0) > 0 \tag{2}$$

Subjective life satisfaction (*SubjLifeSat_{it}*) serves as an observable proxy for utility and is assumed to have a nonlinear relationship with utility. A logarithmic function is used to capture the diminishing marginal utility of satisfaction:

$$SubjLifeSat_{it} = \ln(U_{it} + 1) = \ln(\alpha_0 + \alpha_1 Z_{it} + \alpha_2 IntBankUse_{it} + 1) \tag{3}$$

Within the TAM framework, behavioral intention to use (BI_{it}) serves as a mediating factor for actual usage behavior. It is driven by perceived ease of use (PEU_{it}) and perceived usefulness (PU_{it}), expressed as:

$$BI_{it} = f(PEU_{it}, PU_{it}) \quad (4)$$

This study employs technological well-being ($TechWell_{it}$) as a proxy variable for PEU. Technological well-being is a comprehensive indicator that reflects an individual's ability to access and utilize technological resources, encompassing technology availability, digital literacy, and integrating technology into daily life. PEU refers to the user's perception of the difficulty in using a system or tool, and technological well-being is a fundamental determinant of this perception. Individuals with higher technological well-being are more proficient in applying technology and perceive lower difficulty using Internet banking services. Thus, perceived ease of use can be indirectly measured through individuals' self-assessed technological well-being.

Individuals' e-commerce behavior ($EComm_{it}$) represents PU as a proxy variable. The maturity of e-commerce directly reflects the scale and convenience of commercial activities within the digital economy, encompassing payment method diversity and security. PU refers to users' perception of a system or tool's effectiveness in accomplishing tasks, and the development of e-commerce strengthens users' recognition of the actual value of Internet banking. Additionally, as e-commerce platforms rely on efficient and fast payment methods, they further emphasize the practicality of Internet banking, reinforcing its PU.

Therefore, behavioral intention can be explicitly expressed as:

$$Pr(BI_{it} = 1) = \text{Logit}(\beta_0 + \beta_1 TechWell_{it} + \beta_2 EComm_{it} + \beta_3 Z_{it}) \quad (5)$$

Given the strong functional and goal-oriented correlation between electronic payments and Internet banking services, this study employs electronic payments ($ElecPay_{it}$) as a proxy variable for behavioral intention. Both fall within the domain of digital financial technology applications. The increasing demand for electronic payments is often accompanied by a greater need for more efficient financial services, effectively reflecting users' inclination to adopt Internet banking. Moreover, electronic payments and Internet banking share similar technological attributes, such as operational convenience (PEU) and functional utility (PU). As a result, electronic payment behavior is a reliable indicator of users' behavioral intention, aligning with the core constructs of the TAM.

Since behavioral intention influences the actual usage of Internet banking services, the model can be formulated as follows:

$$Pr(IntBankUse_{it} = 1) = \text{Logit}(\gamma_0 + \gamma_1 BI_{it} + \gamma_2 EComm_{it} + \gamma_3 Z_{it}) \quad (6)$$

The actual usage of Internet banking services ultimately influences individuals' subjective life satisfaction. This effect can be expressed as:

$$SubjLifeSat_{it} = \lambda_0 + \lambda_1 IntBankUse_{it} + \varepsilon_{it} \quad (7)$$

Since subjective life satisfaction is measured using a truncated scale ranging from 1 to 10 in the survey responses, the estimation of Eq. (7) will be conducted using a Tobit model.

Additionally, in 2021, Taiwan launched PIBs, expanding financial services through a more comprehensive and ecosystem-driven approach. To further examine the impact of PIBs introduction on subjective life satisfaction, this study employs the Difference-in-Differences (DID) method, which accounts for time dynamics and differences between the treatment and control groups. The regression equation can be expressed as:

$$SubjLifeSat_{it} = \delta_0 + \delta_1 Post_t + \delta_2 IntBankUse_{it} + \delta_3 (Post_t \times IntBankUse_{it}) + e_{it} \quad (8)$$

Here, $Post_t$ is a post-event dummy variable, and $IntBankUse_{it}$ represents the treatment group dummy variable. The interaction term $Post_t \times IntBankUse_{it}$ captures the net impact of the introduction of PIBs.

The overall model framework is built upon the TAM as its core theoretical foundation. Through mathematical representation and empirical design, it establishes the link between behavioral intention, actual usage behavior, and subjective life satisfaction while capturing the dynamic impact of event intervention. The hierarchical and structural design of the model enables a more comprehensive understanding of the economic and social benefits of technology adoption on life satisfaction.

5. Empirical Results

This study examines data from Taiwan's Digital Development Survey conducted in 2019, 2021, and 2022 to identify key factors influencing Internet banking adoption. Due to the COVID-19 pandemic, the 2020 survey was canceled for the first time since its inception in 2001, making data from that year unavailable. Since 2021 marked the launch of Taiwan's PIBs, this study uses 2019 as the baseline year, with 2021 and 2022 serving as observation periods to represent the launch year and post-launch phase. This design allows for a comprehensive analysis of PIBs' impact on Internet banking usage.

The study first examines the determinants of Internet banking adoption using a Logit model to assess key influencing factors. It then applies the DID method to evaluate whether PIBs introduction significantly affected Internet banking usage, comparing changes from pre-launch (2019) to launch year (2021) and from launch year to post-launch (2022). This structured approach provides a detailed assessment of both individual adoption behavior and the broader impact of digital financial innovation, offering insights into the evolving role of PIBs in Taiwan's financial landscape.

5.1. Empirical results on the determinants and impact of internet banking usage

5.1.1. Empirical results on the determinants of behavioral intention toward electronic payments

According to the TAM, behavioral intention is a key predictor of actual behavior, with PEU and PU as its core determinants. This study uses technological well-being and e-commerce engagement as proxy variables for PEU and PU, respectively, while electronic payment usage is a proxy for behavioral intention. A Logit model is applied to examine the behavioral intentions of Taiwanese individuals from 2019 to 2022. Table 2 presents the empirical results, with electronic payment adoption as the proxy for behavioral intention.

In the pooled data analysis, the marginal effect of technological well-being is 0.0378, meaning that each unit increase in technological well-being raises individuals' behavioral intention to use electronic payments by 3.78 %. This result supports the TAM framework, confirming the positive influence of PEOU on behavioral intention. Similarly, the marginal effect of e-commerce engagement is 29.41 %, indicating widespread e-commerce adoption significantly strengthens behavioral intention. This finding highlights PU's critical role in shaping behavioral intention—when digital tools are accessible and easy to use, and e-commerce offers substantial functional value, individuals are more likely to adopt electronic payments.

Regarding personal characteristics, the results show that each additional year of age lowers the likelihood of electronic payment adoption by 0.17 %. This decline may stem from older adults' challenges with technology adaptation and lower digital literacy. Additionally, men show significantly lower behavioral intention than women, with a marginal effect of -0.0356 . This difference may reflect traditional gender roles, where women are more involved in household purchases, increasing their exposure to digital payment technologies.

Education level plays a crucial role, with each additional year of schooling increasing behavioral intention by 4.48 %. Higher education enhances individuals' technological acceptance and adoption. Internet usage frequency has a strong marginal effect of 24.99 %, suggesting that

Table 2

Empirical results of determinants influencing electronic payment behavioral intentions.

Logit Model	Behavioral Intention to Use: Electronic Payment (ElecPay)			
	2019–2022	Year 2019	Year 2021	Year 2022
Perceived Ease of Use				
Technological Well-being (TechWell)	0.0378*** (21.86)	0.0352*** (13.62)	0.0306*** (9.68)	0.0417*** (12.60)
Perceived Usefulness				
Electronic Commerce (EComm)	0.2941*** (44.07)	0.2455*** (24.23)	0.2967*** (24.26)	0.3317*** (26.44)
Personal Characteristic				
Age (Age)	−0.0017*** (−9.58)	−0.0002 (−1.14)	−0.0032*** (−9.36)	−0.0036*** (−10.00)
Gender (Gender)	−0.0356*** (−6.49)	−0.0076 (−0.95)	−0.0637*** (−6.29)	−0.0513*** (−4.88)
Years of Education (EduYears)	0.0448*** (38.34)	0.0326*** (18.94)	0.0508*** (23.68)	0.0527*** (23.57)
Frequency of Internet Use (FreqNetUse)	0.2499*** (8.78)	0.2215*** (5.30)	0.2613*** (4.93)	0.2527*** (4.74)
Metropolitan Area (MetroArea)	0.0566*** (10.13)	0.0395*** (4.67)	0.0623*** (6.12)	0.0438*** (4.14)
LR χ^2 (p-value)	9682.59*** (0.00)	2695.21*** (0.00)	3380.29*** (0.00)	3665.73*** (0.00)
Number of obs.	33,051	11,708	10,427	10,916

a. Values in parentheses represent t-values. b. All regression models include a constant term. c. *** represent statistical significance at the 1 %.

frequent Internet users are likelier to adopt electronic payments. Urban residents also show a higher behavioral intention than rural residents, with a marginal effect of 0.0566, reflecting disparities in technological resources and digital payment infrastructure.

Examining yearly trends, technological well-being, and e-commerce consistently drive electronic payment adoption from 2019 to 2022. The marginal effect of technological well-being ranges from 3.06 % to 4.17 %, while e-commerce engagement ranges from 24.55 % to 33.17 %, showing a growing influence over time. This trend suggests that as e-commerce expands, its impact on behavioral intention strengthens, reinforcing the roles of PEOU and PU. After Taiwan introduced PIBs in 2021, the influence of PEOU and PU peaked in 2022, indicating that PIBs successfully lowered technological barriers and improved perceived utility. These results suggest that PIBs act as a catalyst for technology acceptance, accelerating the adoption of digital financial services.

The cross-year analysis of personal characteristics reveals changing influences on behavioral intention toward electronic payments. In 2019, the effects of age and gender were relatively weak, whereas education level and internet usage frequency showed significant positive impacts, with marginal effects of 3.26 % and 22.15 %, respectively. Urban residency had a marginal effect of 3.95 %, reflecting regional disparities in digital finance adoption.

By 2021, the negative effects of age and gender became more pronounced, with marginal effects of −0.0032 and −0.0637, respectively. In contrast, education level and internet usage frequency had more substantial positive effects, increasing to 5.08 % and 26.13 %. The influence of urban residency also rose to 6.23 %, suggesting an expanding urban-rural divide in digital financial adoption.

In 2022, these trends continued. The marginal effects of education level and internet usage frequency reached 5.27 % and 25.27 %, reinforcing their stable role in promoting technology acceptance. Meanwhile, the negative effects of age and gender remained significant at −0.0036 and −0.0513. The influence of urban residency slightly declined to 4.38 %, possibly indicating improved digital accessibility in

non-urban areas. Personal characteristics are crucial determinants of behavioral intention, with education and digital engagement playing key roles in technology adoption. However, the persistent negative effects of age and gender highlight significant barriers, underscoring the need for targeted strategies to bridge the digital divide.

5.1.2. Empirical results on the determinants of actual internet banking usage

Building on the previous analysis of behavioral intention, this study uses Internet banking usage as a proxy for actual usage behavior. A Logit model is applied to examine the determinants of usage among Taiwanese individuals from 2019 to 2022, focusing on the effects of behavioral intention, perceived usefulness, and personal characteristics.

The empirical results in Table 3 show that behavioral intention has a strong positive effect on actual usage, with a marginal effect of 0.8698. This suggests that a 1 % increase in the probability of engaging in electronic payment behavior raises the probability of using Internet banking services by 86.98 %. These findings empirically confirm the core principle of the TAM—that behavioral intention is the primary driver of actual behavior. Additionally, e-commerce significantly positively affects Internet banking usage, with a marginal effect of 13.17 %. This indicates that as e-commerce expands and matures, it plays a crucial role in promoting Internet banking adoption. The growing reliance on digital financial services is closely linked to the continuous development of e-commerce, reinforcing its importance in shaping consumer financial behavior.

Regarding personal characteristics, education level, and internet usage frequency exhibit significant and stable positive effects on Internet banking usage, with marginal effects of 4.15 % and 24.14 %, respectively. These results indicate that individuals with higher education levels and frequent Internet use are more likely to adopt and actively use Internet banking services. Additionally, urban residency has

Table 3

Empirical Results of Behavioral Determinants Influencing Internet Banking Services.

Logit Model	Actual Usage Behavior: Internet Banking Usage Behavior (IntBankUse)			
	2019–2022	Year 2019	Year 2021	Year 2022
Behavioral Intention to Use				
Electronic Payment (ElecPay)	0.8698*** (18.98)	0.9060*** (11.79)	0.7292*** (8.10)	0.8323*** (10.53)
Perceived Usefulness				
Electronic Commerce (EComm)	0.1317*** (8.61)	0.1058*** (4.47)	0.1749*** (6.12)	0.1520*** (5.47)
Personal Characteristic				
Age (Age)	−0.0002 (−0.87)	0.0003 (1.06)	−0.0007 (−1.43)	−0.0009* (−1.82)
Gender (Gender)	−0.0248*** (−3.58)	−0.0477*** (−4.32)	−0.0204 (−1.55)	−0.0128 (−1.04)
Years of Education (EduYears)	0.0415*** (17.84)	0.0458*** (13.27)	0.0453*** (9.49)	0.0385*** (9.18)
Frequency of Internet Use (FreqNetUse)	0.2414*** (8.22)	0.2262*** (4.91)	0.2934*** (5.21)	0.2300*** (4.51)
Metropolitan Area (MetroArea)	0.0318*** (4.33)	0.0376*** (3.11)	0.0317** (2.41)	0.0357*** (2.86)
LR χ^2 (p-value)	12,475.36*** (0.00)	4001.19*** (0.00)	4078.73*** (0.00)	4391.09*** (0.00)
Number of obs.	33,051	11,708	10,427	10,916

a. Values in parentheses represent t-values. b. All regression models include a constant term. c. ***, ** and * represent statistical significance at the 1 %, 5 % and 10 %.

a marginal effect of 3.18 %, emphasizing the role of urbanization in shaping technology adoption. The marginal effect of gender is −2.48 %, suggesting that women are significantly more likely than men to use Internet banking services. However, age does not have a statistically significant effect on actual usage behavior, implying that while age influences behavioral intention, its direct impact on actual adoption is limited.

Examining the yearly results, behavioral intention had the most substantial effect on Internet banking usage in 2019, with a marginal effect of 90.60 %. This suggests that before PIBs emerged, behavioral intention was the primary driver of adoption. The marginal effect of e-commerce was 10.58 %, indicating that its influence on Internet banking usage was still developing. Education level and internet usage frequency exhibited marginal effects of 4.58 % and 22.62 %, respectively, highlighting the importance of digital literacy and engagement. Urban residency had a marginal effect of 3.76 %, reflecting a persistent urban-rural gap, where individuals in metropolitan areas were significantly more likely to adopt Internet banking services than those in rural regions.

The 2021 and 2022 results highlight the evolving influence of behavioral intention and e-commerce on Internet banking usage. In 2021, the marginal effect of behavioral intention was 72.92 %, lower than in 2019, but it remained the primary driver of actual usage behavior. Meanwhile, e-commerce exhibited a marginal effect of 17.49 %, reflecting its growing maturity and more substantial influence on adoption. Education level and internet usage frequency continued to have strong positive effects, with marginal effects of 4.53 % and 29.34 %, respectively.

By 2022, the marginal effect of behavioral intention rebounded to 83.23 %, reaffirming its critical role in actual usage. The effect of e-commerce slightly declined to 15.20 %, though it remained statistically significant. Education level and internet usage frequency sustained stable positive effects, with marginal effects of 3.85 % and 23.00 %, respectively. Urban residency's impact slightly increased to 3.57 %, reflecting a continued urban advantage in Internet banking adoption.

The empirical results from 2019 to 2022 consistently demonstrate that behavioral intention is the primary driver of Internet banking usage across all years. While its influence fluctuated, it remained stable and statistically significant. The expansion of e-commerce played a crucial role in promoting adoption, with its peak influence in 2021. The steady effects of education level and internet usage frequency further emphasize the importance of digital literacy and online engagement in fostering technology acceptance. However, persistent gender disparities and urban-rural gaps indicate structural differences in technology adoption. Policymakers should address these gaps by promoting digital financial inclusion, expanding access to technology, and strengthening financial literacy programs to support broader adoption of Internet banking services across diverse demographic groups.

5.1.3. Analysis of the impact of internet banking usage on subjective life satisfaction

Table 4 presents the empirical results on the impact of Internet banking usage on individuals' subjective life satisfaction. The data exhibits censoring characteristics since subjective life satisfaction is measured on a 1 to 10 scale in the survey responses. Given that the objective of this study is to estimate the potential effect of Internet banking usage on subjective life satisfaction, the Tobit model is employed for analysis to account for the censored nature of the dependent variable.

The pooled data analysis shows that Internet banking usage significantly positively affects subjective life satisfaction, with a marginal effect of 0.7076. This suggests that for each 1 % increase in the probability of using Internet banking, subjective life satisfaction improves by an average of 0.7076 points. These findings indicate that Internet banking enhances transaction efficiency and financial management while improving overall quality of life. By reducing transaction time, lowering

Table 4

Empirical Results of the Impact of Internet Banking Service on Subjective Life Satisfaction.

Tobit Model Actual Usage Behavior: Internet Banking Usage Behavior (IntBankUse)	Subjective Life Satisfaction (SubLifeSat)		
	Marginal Effect	LR χ^2 (p-value)	Number of obs.
2019–2022	0.7076*** (22.28)	492.59*** (0.00)	33,051
Year 2019	1.1162*** (19.81)	385.87*** (0.00)	11,708
Year 2021	0.3876*** (6.99)	48.75*** (0.00)	10,427
Year 2022	0.5491*** (10.29)	105.35*** (0.00)	10,916

a. Values in parentheses represent t-values. b. All regression models include a constant term. c. *** represent statistical significance at the 1 %.

operational costs, and offering diverse financial services, Internet banking contributes to higher life satisfaction. The positive effect underscores its role as a modern financial tool, particularly in digital economic growth. Its efficiency and convenience drive improved living standards, reinforcing the broader societal benefits of financial technology adoption.

Yearly analysis reveals that the marginal effect was highest in 2019, at 1.1162, suggesting that Internet banking had the most pronounced impact on life satisfaction during this period. This may reflect the growing role of Internet banking in providing a seamless financial experience and enhancing daily convenience, mainly as its functionalities mature. The increasing efficiency and accessibility of Internet banking in 2019 likely contributed to its more substantial effect on life satisfaction.

In 2021, the marginal effect dropped to 0.3876, possibly due to the introduction of PIBs. As a new financial service model, PIBs likely required an adaptation period before their full impact on life satisfaction became evident. Market adjustments and initial consumer hesitation may have temporarily reduced the immediate benefits of Internet banking. By 2022, the marginal effect rebounded to 0.5491, suggesting that as PIBs services expanded and their functionalities improved, users became more familiar with traditional Internet banking and PIBs options. Enhanced accessibility, efficiency, and financial management capabilities contributed to a steady increase in life satisfaction.

These findings highlight the policy implications of PIBs and their influence on subjective life satisfaction. By applying a Difference-in-Differences (DID) model, this study isolates the dynamic effects of PIBs introduction across different years. The DID method controls for unobserved year-specific fixed effects, allowing for a more precise estimation of the impact of PIBs on Internet banking usage and subjective life satisfaction before and after their market entry. This approach provides a comprehensive evaluation of both the short-term and long-term effects of PIBs, offering valuable insights into their social benefits and implications for digital financial policies.

5.2. Analysis of the impact of purely internet-based banks on subjective life satisfaction under the difference-in-differences approach

Table 5 presents the results of the DID estimation on the impact of PIBs introduction on individuals' subjective life satisfaction. The findings reveal the dynamic effects of PIBs across different years. In 2021, when PIBs were officially launched, life satisfaction among users increased from 7.087 in 2019 to 7.250, while non-user satisfaction rose from 6.893 to 7.229. However, the gap between the two groups narrowed significantly to 0.021, suggesting that in the early phase of PIBs adoption, the improvement in life satisfaction for users was relatively modest. This implies that the full potential of the new banking model

Table 5
DID Results of the Impact of Pure Internet Banks Launch on Subjective Life Satisfaction.

DID		Subjective Life Satisfaction	
Year 2019 (Pre-PIBs)	Treatment	7.087	Difference
	Control	6.893	0.194(5.64)***
Year 2021 (Launch Year- PIBs)	Treatment	7.250	Difference
	Control	7.229	0.021(0.74)***
		Difference-in-Differences (DID)	-0.173(3.78)***
Year 2021 (Launch Year- PIBs)	Treatment	7.110	Difference
	Control	7.087	0.022 (0.67)
Year 2022 (Post 1 Year- PIBs)	Treatment	7.276	Difference
	Control	7.250	0.026 (0.88)
		Difference-in-Differences (DID)	0.003 (0.08)

a. Values in parentheses represent t-values. b. *** represent statistical significance at the 1 %.

had not yet materialized, and its impact on life satisfaction was more pronounced for non-users than expected.

The DID analysis results indicate that PIBs introduction had a negative and statistically significant impact on subjective life satisfaction, with a marginal effect of -0.173 . This suggests that the initial phase of PIBs operations encountered several challenges. The services offered by PIBs may not have fully met user expectations. Despite their convenience, PIBs may have lacked traditional banks' comprehensive features or user experience. Additionally, users likely faced adaptation costs, including learning how to navigate Internet banking platforms, adjusting to the absence of physical branches, and building trust in the newly established PIBs model. These factors may have contributed to the initial decline in life satisfaction among PIBs users.

Two of Taiwan's largest PIBs are Japanese and Korean banking group subsidiaries, which may have added to the initial challenges. Many Taiwanese consumers had no prior exposure to financial services, leading to skepticism or hesitation in adopting these platforms. Additionally, the entry of PIBs may have diverted customers from traditional online banks, causing dissatisfaction with the overall online banking experience. This shift in customer behavior could explain the initial decline in overall satisfaction with Internet banking services.

By 2022, PIBs had improved their services after one year of operation, and user adaptation had significantly increased. As a result, users' subjective life satisfaction rose to 7.276, while non-users' satisfaction increased to 7.250. Although the gap between the two groups widened, the DID analysis shows that the impact of PIBs on subjective life satisfaction became neutral, with a marginal effect of 0.003, which is not statistically significant. This suggests that as PIBs services improved, users gradually accepted their convenience and efficiency. However, these benefits had not yet translated into a measurable increase in subjective life satisfaction.

Several factors could explain this neutral impact in 2022. First, users may have had high expectations for PIBs regarding transaction convenience, service diversity, and efficiency. However, the actual performance of these services in the early stages may not have fully met these expectations. Technical maturity, system stability, and service scope were still developing. Second, PIBs require time to integrate with other financial platforms and build a supporting ecosystem through complementary policies and regulations. This stabilization process delayed any immediate improvements in life satisfaction. In other words, PIBs in their early phase focused more on structural adjustments than immediate user benefits, with potential positive effects likely to accumulate over time as services evolve and integrate into the broader financial system.

Compared to the Tobit model results in Section 5.1.3, which indicate a positive and significant effect of Internet banking on subjective life satisfaction—particularly in 2019—the DID model provides deeper insight into the dynamic impact of PIBs. Before their launch, traditional online banking had a stable, positive effect on life satisfaction. However,

introducing PIBs initially led to short-term negative effects, which gradually stabilized as services matured. These findings suggest that PIBs, as a new form of financial service, should prioritize user education and functional enhancements during their early stages to reduce adaptation costs and maximize long-term benefits.

By combining the Tobit and DID model results, this study provides a more comprehensive assessment of online banking services' short- and long-term effects on subjective life satisfaction. The findings offer empirical evidence for future digital finance policy development, highlighting the importance of gradual service improvements, regulatory support, and strategies to enhance consumer trust and usability in new financial service models.

6. Discussion and conclusion

6.1. Discussion

This study analyzes official digital survey data from Taiwan's Ministry of Digital Affairs, integrating the Technology Acceptance Model and the Theory of Behavioral Utility. It employs a Difference-in-Differences research design to examine key factors influencing Taiwanese citizens' use of Internet banking services and assess the impact of Pure Internet Banks' (PIBs) market entry. The empirical results show that perceived usefulness and ease of use are the main determinants of behavioral intention. E-commerce usage frequency and technological well-being are proxies for these factors, highlighting the connection between consumers' acceptance of digital financial tools, their digital adaptability, and the broader digital environment.

Electronic payment behavior plays a key role in Internet banking usage, showing that reliance on electronic payments has significantly driven adoption. The study also finds a positive relationship between years of education, Internet usage frequency, and the adoption of digital financial services, stressing the importance of digital literacy and education in technology acceptance. In contrast, age and gender negatively affect behavioral intention, suggesting that older individuals face more significant challenges adapting to digital financial services. Women, however, show a higher tendency to use electronic payment tools, possibly due to differences in consumption habits and financial management roles.

Regarding PIBs' market entry, their introduction in 2021 significantly increased behavioral intention toward Internet banking. However, in the initial adaptation phase, subjective life satisfaction temporarily declined, likely due to the learning costs of new technologies. By 2022, this negative effect was no longer significant, indicating that as consumers became more familiar with PIBs services, their satisfaction levels stabilized, though no noticeable growth followed. These findings confirm the applicability of TAM in digital finance and provide insights into how PIBs influence consumer behavior and life quality over time.

The findings of this study align with prior research on digital finance and technology acceptance while providing a dynamic perspective on the market impact of PIBs. The analysis of PU and PEOU confirms the conclusions of Liébana-Cabanillas et al. (2018) and Bektukhambetova & Németh (2023), emphasizing the role of technological adaptability and the digital application environment in shaping technology acceptance. Additionally, the results support Chandio et al. (2013), reinforcing the link between user confidence in technology and adoption behavior.

Regarding individual characteristics, this study confirms the interaction between age, education level, and digital behavior, consistent with Grigorescu et al. (2023), who found that individuals with higher education levels and frequent digital tool usage are more likely to adopt new financial technologies. The observed gender-based differences also support Yu (2012), showing that women are more accepting of electronic payment services, likely due to differences in financial management roles and spending habits.

For the short- and long-term effects of PIBs' market entry, the DID analysis shows that PIBs initially had a negative impact on subjective life satisfaction, which later stabilized. This partially echoes [Furst et al. \(2002\)](#), who studied the U.S. electronic banking market and aligns with [Ren et al. \(2022\)](#), who highlighted the volatility of digital economic development. These findings confirm that Internet banking reforms do not yield immediate, stable benefits but require time for adaptation to realize their full social and economic impact.

From the perspective of TAM and behavioral economics, this study provides strong evidence that technological well-being influences PEOU, enhancing behavioral intention and actual usage. The results highlight the critical role of digital skills in technology acceptance. Individuals with higher technological well-being perceive digital financial tools as more accessible and are more likely to adopt them. Additionally, the reinforcing effect of e-commerce behavior on PU explains why consumers widely accept electronic payments and Internet banking, as frequent e-commerce use strengthens the perceived value of digital financial services.

Following the introduction of PIBs, the market adaptation effect aligns with previous research. Initially, consumers experienced a decline in satisfaction due to concerns about security, risks, and operational uncertainties. However, as users became more familiar with PIBs and learning costs decreased, this negative impact gradually diminished. This process supports [Sharma and Sharma \(2019\)](#), who highlighted the influence of trust and security risks on the willingness to use PIBs, reinforcing the importance of market adjustment and event intervention in developing PIBs.

This study provides valuable insights for government regulators, financial institutions, and consumer protection policymakers. The government should enhance digital financial education and promote digital inclusion, especially for older adults and individuals with lower education levels. Initiatives such as digital literacy programs and simplified digital financial tools can help bridge the technology gap, increasing trust and adoption of online banking. The negative effects observed during PIBs' adaptation phase highlight the need for a well-planned market entry strategy. Measures such as waiving transaction fees during the trial period, introducing phased access to high-risk transactions, and providing dedicated customer support for real-time assistance and user education can help lower learning costs and stabilize user satisfaction.

Building trust in digital finance requires strong data security and privacy protection. Strengthening security measures, implementing multi-factor authentication, promoting privacy-enhancing technologies, and ensuring transparent risk disclosure are essential for increasing consumer confidence. Requiring PIBs to commit to blockchain-based transaction transparency and cybersecurity compliance, alongside periodic government-led disclosures of regulatory adherence and user satisfaction rankings, could foster healthier competition and stronger consumer protection.

This study has certain limitations, which also suggest avenues for future research. For instance, the data span (2019, 2021, and 2022) captures only the immediate, short-term effects of PIBs' introduction without observing longer-term impacts beyond 2023. Future studies should extend the observation period to evaluate how PIBs influence consumer behavior and life satisfaction as the market matures. In addition, while convenient, the use of electronic payment behavior as a proxy for Internet banking intention does not fully reflect the range of online banking activities (e.g., fund transfers, investment management, loan applications). Future research could incorporate more granular transaction data or multiple usage indicators to provide a more comprehensive view of digital financial behaviors across services.

Moreover, our analysis did not explicitly account for external factors such as financial market fluctuations, macroeconomic conditions, and regulatory changes. Future investigations might employ methods like structural equation modeling or machine learning to include these variables, thereby strengthening causal inference and the robustness of

results. Another limitation is the focus on Taiwan's market, which may limit the generalizability of the findings to other contexts. Comparative studies spanning other digitally advanced economies (for example, South Korea, Japan, or China) could enhance external validity by revealing how different institutional and cultural settings moderate the impact of PIBs on consumers.

Finally, as digital finance evolves, future work could explore the interplay between PIBs and emerging technologies—such as AI-driven financial advisory services and blockchain-based payment systems—to understand their combined effects on Internet banking adoption and user satisfaction. By addressing these limitations and pursuing such directions, subsequent studies can build on our findings and further illuminate digital finance's role in consumer well-being.

6.2. Conclusion

This study analyzes official digital survey data from Taiwan's Ministry of Digital Affairs, integrating the Technology Acceptance Model and the Theory of Behavioral Utility. Using the Difference-in-Differences method, it examines key factors influencing Taiwanese citizens' adoption of Internet banking and assesses the market impact of Pure Internet Banks (PIBs). The empirical results show that perceived usefulness (PU) and perceived ease of use (PEOU) are critical determinants of behavioral intention toward electronic payments. Technological well-being significantly enhances PEOU, while e-commerce participation strongly correlates with PU, highlighting the role of digital environments and usage habits in Internet banking adoption. The study also confirms that behavioral intention directly influences actual Internet banking usage, reinforcing the applicability of TAM in digital finance.

Regarding personal characteristics, education level, and Internet usage frequency positively influence technology adoption, emphasizing the importance of digital literacy. In contrast, age and gender negatively impact adoption, suggesting that older individuals face more significant challenges adapting to digital financial services. At the same time, female users tend to adopt electronic payments and Internet banking more than males.

For the short- and long-term effects of PIBs, their introduction in 2021 initially led to a decline in subjective life satisfaction, likely reflecting an adaptation period as the market adjusted to the new service. By 2022, this negative sentiment had stabilized, but no significant positive effects had emerged. These findings validate TAM's explanatory power in Internet banking adoption and provide empirical insights into how PIBs influence consumer behavior and subjective life satisfaction over time.

Theoretically, this study extends TAM by incorporating digital environmental factors, such as technological well-being and e-commerce participation, into the model. This addresses gaps in existing research on how digital ecosystems shape technology acceptance. Additionally, by integrating the DID model, the study evaluates PIBs' event impact from a dynamic perspective, contributing to the discussion on the short- and long-term effects of digital financial development. The research also demonstrates TAM's applicability across demographic groups, providing evidence on the interactive effects of age, gender, education level, and digital usage habits. These findings deepen the understanding of digital inequality and its implications for financial technology adoption. The study enhances research on digital financial inclusion and consumer behavior by linking technology acceptance theory with policy evaluation.

From a practical perspective, the findings offer insights for government regulators, financial institutions, and consumer protection policymakers, particularly in reducing the digital divide and improving financial inclusion. Governments and financial institutions should collaborate to enhance digital financial education and introduce simplified services that lower barriers for older adults and individuals with lower educational attainment.

Special attention should be given to the "adaptation period effect"

observed in PIBs' early phase. To reduce user uncertainty and shorten learning curves, policymakers and financial institutions should implement strategies such as offering trial access to low-risk financial products, providing dedicated professional customer support, and ensuring transparent risk disclosure. These measures can help build consumer confidence in Internet banking services. Strengthening security and trust in digital finance is also essential. Financial institutions should integrate blockchain technology, adopt zero-trust architectures, and enhance multi-factor authentication to improve data security and fraud prevention. Meanwhile, the government should establish transparency indicators for digital finance and regularly publish compliance reports and bank consumer satisfaction rankings. These efforts would enhance market confidence and contribute to a stable and trustworthy digital financial ecosystem.

This study confirms the strong explanatory power of TAM in digital finance. It highlights the challenges and opportunities PIBs face during market entry and offers practical recommendations for policymakers and financial institutions. These contributions provide a valuable reference for efforts to close the digital divide and enhance consumers' life satisfaction in an era of ongoing digital transformation.

Compliance with ethical standards

Disclosure of potential conflicts of interest

The author declares that they have no conflicts of interest to disclose.

Research involving human participants and/or animals

This study did not involve any research with human participants or animals.

Informed consent

Not applicable, as this study did not involve human participants.

Research design

This study adopts a quantitative research design, utilizing a combination of econometric models to analyze the determinants of Internet banking adoption and its impact on subjective life satisfaction. The study leverages nationally representative data from Taiwan's Digital Development Survey (2019, 2021, and 2022) conducted by the Ministry of Digital Affairs. Three econometric models are employed: (1) a Logit model to identify key determinants influencing Internet banking adoption, (2) a Tobit model to examine the relationship between Internet banking usage and subjective life satisfaction, and (3) a Difference-in-Differences (DID) approach to evaluate the policy impact of purely Internet-based banks (PIBs) on consumer behavior and satisfaction. The study integrates the Technology Acceptance Model (TAM) as the theoretical framework to analyze consumer decision-making in the digital financial landscape. The findings provide empirical insights into technological well-being, e-commerce participation, and digital finance in shaping financial inclusion and consumer well-being.

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CRedit authorship contribution statement

Kuang-Hsien Wang: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declares that he has no competing interests related to this research.

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Data availability statement

The data that support the findings of this study are openly available in the Survey Research Data Archive of Academia Sinica, conducted by the Ministry of Digital Affairs of the Taiwan government, at the following links:

<https://doi.org/10.6141/TW-SRDA-AE010018-1>

<https://doi.org/10.6141/TW-SRDA-AE010019-1>

<https://doi.org/10.6141/TW-SRDA-AE010020-1>

Data availability

The authors do not have permission to share data.

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